

Bias and variance of a success-weighted subset

Asymptotic research note; independently reviewed

30 September 2026

This finite algebraic lemma controls the averaging error in a forced-success polarization argument. It is independent of the comparison-moment identities.

Lemma 1. *Let $p_1, \dots, p_m \in (0, 1]$, $1 \leq \ell < m$, $h = m - \ell$, and choose an ℓ -element subset B with probability*

$$\Pr(B) = \frac{\prod_{j \in B} p_j}{e_\ell(p)}.$$

For any $a > 0$, put $x_j = a(1 - p_j)$, $\bar{x} = m^{-1} \sum_j x_j$, $x_R = h^{-1} \sum_{j \notin B} x_j$, and $\Delta = x_R - \bar{x}$. Then

$$\mathbb{E}\Delta = \frac{a}{mh} \sum_{i < j} (p_i - p_j)^2 \frac{e_{\ell-1}(p_{-[i,j]})}{e_\ell(p)} \geq 0. \quad (1)$$

If $b = \min_i p_i$ and $v = \sum_i (p_i - \bar{p})^2$, then

$$0 \leq \mathbb{E}\Delta \leq \frac{av}{hb}, \quad \text{Var } \Delta \leq \frac{a^2}{h^2} \sum_i (1 - p_i)^2. \quad (2)$$

In particular, writing $\delta = \max_i (1 - p_i)$,

$$\mathbb{E}\Delta^2 \leq \frac{a^2 m \delta^2}{h^2} + \frac{a^2 m^2 \delta^4}{h^2 b^2}. \quad (3)$$

Proof. Let $I_i = \mathbf{1}_{i \in B}$ and $\pi_i = \mathbb{E}I_i$. The two inclusion probabilities have difference

$$\pi_i - \pi_j = (p_i - p_j) \frac{e_{\ell-1}(p_{-[i,j]})}{e_\ell(p)}.$$

Also $\sum_i \pi_i = \ell$, and therefore

$$\sum_i x_i(\pi_i - \ell/m) = \frac{1}{m} \sum_{i < j} (x_i - x_j)(\pi_i - \pi_j).$$

Since $\Delta = -(\sum_i x_i I_i - \ell \bar{x})/h$ and $x_i - x_j = -a(p_i - p_j)$, this proves (1). The ratio in (1) is at most $1/p_i \leq 1/b$, since $e_\ell(p) \geq p_i e_{\ell-1}(p_{-[i,j]})$. Using $\sum_{i < j} (p_i - p_j)^2 = mv$ proves its upper bound.

For the variance, pairwise inclusion covariances are nonpositive. Here is a direct verification, requiring no general negative association theorem. With $E_t = e_{\ell-t}(p_{-[i,j]})$ and $Z = e_\ell(p) = E_0 + (p_i + p_j)E_1 + p_i p_j E_2$, one has

$$\text{Cov}(I_i, I_j) = \frac{p_i p_j (E_0 E_2 - E_1^2)}{Z^2} \leq 0.$$

The final inequality is Newton's elementary-symmetric inequality; indices outside the available range are interpreted as zero. Because $x_i \geq 0$, it follows that

$$\text{Var}\left(\sum_i x_i I_i\right) \leq \sum_i x_i^2 \text{Var}(I_i) \leq \sum_i x_i^2.$$

This proves the variance bound in (2). Finally $v \leq m\delta^2$ and $\sum_i (1 - p_i)^2 \leq m\delta^2$, giving (3). $\square$

Corollary 1 (Application to endpoint comparison rows). *Assume the comparison-tensor hypotheses of `finite_endpoint_odds_balance.tex`, $m \geq 256$, and $s = \sum_i (1 - p_i) \leq m/256$. Let $1 \leq \ell \leq m/2$ and take $a = \ell$ in Lemma 1. There is an absolute constant C such that*

$$0 \leq \mathbb{E}\Delta \leq \frac{C(s+1)^2}{m}, \quad \mathbb{E}\Delta^2 \leq C \left(\frac{(s+1)^2}{m} + \frac{(s+1)^4}{m^2} \right). \quad (4)$$

For every twice continuously differentiable function P on $[0, \ell s/h]$,

$$|\mathbb{E}P(x_R) - P(\bar{x})| \leq |P'(\bar{x})| \frac{C(s+1)^2}{m} + \frac{C}{2} \sup_{0 \leq x \leq \ell s/h} |P''(x)| \left(\frac{(s+1)^2}{m} + \frac{(s+1)^4}{m^2} \right). \quad (5)$$

Proof. The cited finite balance theorem gives $p_i > 0$ and $r_i \leq 128(s+1)/m \leq 1$, hence $b \geq 1/2$ and $\delta \leq 128(s+1)/m$. Since $\ell \leq h$ and $h \geq m/2$, (2)–(3) imply (4). Both x_R and $\bar{x}$ belong to the stated interval, so Taylor's theorem and (4) give (5). $\square$

Corollary 2 (A refinement that vanishes at zero deficit). *In the setting of Corollary 1,*

$$0 \leq \mathbb{E}\Delta \leq \frac{Cs(s+1)}{m}, \quad \mathbb{E}\Delta^2 \leq C \left(\frac{s(s+1)}{m} + \frac{s^2(s+1)^2}{m^2} \right). \quad (6)$$

Proof. In (2), improve both quadratic sums by

$$v \leq \sum_i (1 - p_i)^2 \leq \delta s \leq \frac{128s(s+1)}{m}.$$

The same bounds on ℓ/h and b then prove (6). $\square$

Lemma 2 (Taylor's remainder with a singular derivative envelope). *Let $x > 0$, $v \geq 0$, and let P be twice continuously differentiable on the interval with endpoints x, v . Suppose*

$$|P''(z)| \leq Me^{\gamma z}/z \quad (z > 0)$$

there, with $\gamma \geq 0$. Then

$$|P(v) - P(x) - P'(x)(v - x)| \leq Me^{\gamma \max(x,v)}(v - x)^2/x. \quad (7)$$

Proof. Set $\Delta = v - x$. The integral form of Taylor's theorem uses $\Delta^2 \int_0^1 (1 - t)P''(x + t\Delta) dt$. If $\Delta \geq 0$, $(1 - t)/(x + t\Delta) \leq 1/x$. If $\Delta < 0$, then $x + t\Delta = (1 - t)x + tv \geq (1 - t)x$, giving the same inequality for $t < 1$. Integration proves (7), including $v = 0$; the apparent singularity at the final integration endpoint is canceled by $1 - t$. $\square$

Corollary 3 (The success-weighted mixture costs order d/m). *Retain the comparison-tensor assumptions, take $\ell = \lfloor m/4 \rfloor$, $\alpha = \ell/m$, $x = \alpha s$, and $\theta = 1/16$. Suppose a polynomial P satisfies, for $z > 0$,*

$$|P'(z)| \leq C_0 B \sqrt{d/z} e^{\theta z}, \quad |P''(z)| \leq C_0 B (d/z) e^{\theta z}, \quad (8)$$

where $d \geq 1$, $B \geq 0$. On the rows $s \leq m/256$,

$$(\ell + 1) \mathbb{E}_w [\bar{A} |\mathbb{E}_{B \text{ weighted}} P(x_R) - P(x)| \mathbf{1}_{s \leq m/256}] \leq \frac{CBd}{m}, \quad (9)$$

where the constant depends only on C_0 .

Proof. If $s = 0$, every x_j and Δ is zero, so the error vanishes. For $s > 0$, use the mean term from (8) and the remainder bound (7). Since $0 \leq x, x_R \leq \ell s/h$ and $x = \alpha s$, (6) gives

$$|\mathbb{E}P(x_R) - P(x)| \leq CB\sqrt{d}e^{\theta x} \frac{\sqrt{s}(s+1)}{m} + CBde^{\theta \ell s/h} \left(\frac{s+1}{m} + \frac{s(s+1)^2}{m^2} \right).$$

Maclaurin's inequality gives $\bar{A} \leq e^{-\alpha s}$. The fixed choice of θ leaves a strictly negative exponential factor after multiplication. The row-mass estimate in `finite_endpoint_odds_balance.tex` implies $\mathbb{E}_w(s+1)^k e^{-cs} \leq C_{c,k}/m$ for each fixed $c > 0, k$. Apply it to the displayed terms, multiply by $\ell + 1 = O(m)$, and use $\sqrt{d} \leq d$. This proves (9). $\square$

When averaging over uniformly chosen subsets, let $\bar{A} = \binom{m}{\ell}^{-1} e_\ell(p)$ and $A_B = \prod_{j \in B} p_j$. Then the left side of (5), multiplied by $\bar{A}$, is exactly

$$|\mathbb{E}_{B \text{ uniform}}[A_B P(x_R)] - \bar{A} P(\bar{x})|.$$

Thus the first-order error is a bias of order $1/m$, not the order $1/\sqrt{m}$ obtained by Cauchy–Schwarz. The note alone does not assert a new bound on the number of dice faces; it supplies one component of a proposed structured polynomial argument.